What is a trace map?

A first–principles picture, in the language of categories

Goal: build the idea of a "trace" up from nothing, until the field trace used in this project falls out as one special case.

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1. The everyday trace

For a square matrix, the trace is the sum of the diagonal entries.

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \operatorname{tr} A = a + d.$$

One fact makes the trace special: **it does not depend on the basis you chose.** If you change coordinates, A becomes PAP^{-1}, but

$$\operatorname{tr}(PAP^{-1}) = \operatorname{tr}(A).$$

So the trace is really attached to the *linear map*, not to any matrix picture of it. Anything that is independent of arbitrary choices is a hint that a clean, choice–free (i.e. *categorical*) description is hiding underneath.

2. What a category gives us

A category is just *objects* with *arrows* (maps) between them that you can compose. Our running example:

objects: vector spaces U, V, W -------- arrows: linear maps $f\colon V \to W$

To talk about trace we need two extra pieces of structure that vector spaces have and that we can describe with arrows alone.

(a) A tensor product $\otimes$ and a unit object I. You can glue two spaces into $V \otimes W$, and there is a trivial one–dimensional space $I = \Bbbk$ (the ground field) that acts as "nothing added": $I \otimes V \cong V \cong V \otimes I$. A category with such a well–behaved $\otimes$ and unit I is called *monoidal.*

(b) A dual V^*. Every finite–dimensional V has a dual V^* (the linear functionals on V), and it comes with two canonical arrows:

$$I \xrightarrow{\quad\eta\quad} V \otimes V^* \qquad\qquad V^* \otimes V \xrightarrow{\quad\varepsilon\quad} I$$

- **Evaluation** $\varepsilon\colon V^* \otimes V \to I$ "uses" a functional on a vector: $\varepsilon(\phi \otimes v) = \phi(v)$.
- **Coevaluation** $\eta\colon I \to V \otimes V^*$ is the basis–free name for the identity matrix: $\eta(1) = \sum_i e_i \otimes e_i^*$.

An object with such a partner V^*, ε, η is called **dualizable**. Finite–dimensional spaces are exactly the dualizable ones.

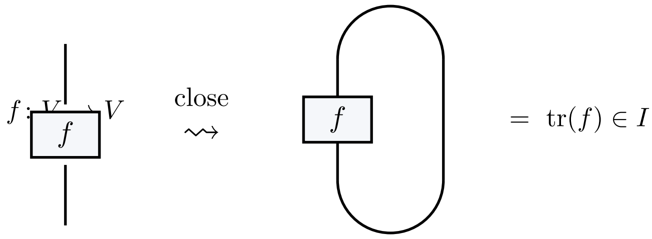
3. Trace as a single loop

Here is the whole idea in one line. Take an endomorphism $f\colon V \to V$ (a map from a space to *itself*). Feed the unit I through this circuit:

$$I \xrightarrow{\ \eta\ } V \otimes V^* \xrightarrow{f \otimes \mathrm{id}} V \otimes V^* \xrightarrow{\text{swap}} V^* \otimes V \xrightarrow{\ \varepsilon\ } I$$

The composite $I \to I$ is just multiplication by a number. **That number is the trace of f**, written $\mathrm{tr}(f)$. No basis was ever mentioned.

The same thing as a picture (string diagram). Read time from bottom to top. A wire is the object V; the cap is η, the cup is ε; the box is f. Tracing = "close the wire into a loop."



In coordinates this loop computes $\sum_i e_i^*\big(f(e_i)\big) = \sum_i A_{ii}$, recovering "sum of the diagonal." The categorical definition just says the same thing without ever picking the e_i.

Why this is the "right" definition. It works in *any* symmetric monoidal category with duals, not only vector spaces, and it automatically obeys the properties we expect:

$$\mathrm{tr}(g \circ f) = \mathrm{tr}(f \circ g), \qquad \mathrm{tr}(\mathrm{id}_V) = \dim V, \qquad \mathrm{tr}(f \oplus g) = \mathrm{tr}\, f + \mathrm{tr}\, g.$$

These are now theorems about the loop, provable by sliding boxes along the wire.

4. From "trace of a map" to "trace map of a field"

So far tr eats one endomorphism and returns a number. The **trace map** of a field extension is what you get when you have a *whole family* of endomorphisms, one for every element of a bigger field.

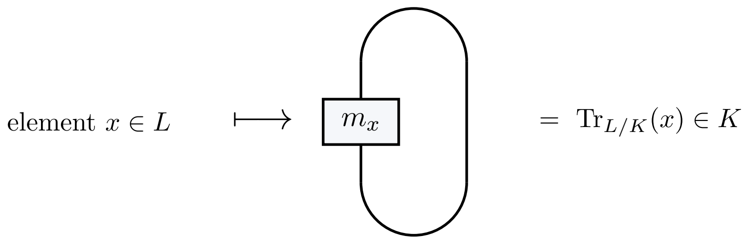
Let $K \subseteq L$ be a field extension. Viewed with K as scalars, L is a K–vector space (finite–dimensional). Each $x \in L$ gives an endomorphism

$$m_x\colon L \to L, \qquad m_x(y) = x \cdot y \quad (\text{“multiply by } x\text{”}),$$

which is K–linear. Now apply the loop of §3 to every m_x at once:

$$L \xrightarrow{\ x \mapsto m_x\ } \{K\text{-endomorphisms of } L\} \xrightarrow{\text{tr (§3 loop)}} K$$

The composite $x \mapsto \mathrm{tr}(m_x)$ is the **field trace** $\mathrm{Tr}_{L/K}\colon L \to K$. It is K–linear because tr is additive and $m_{x+x'} = m_x + m_{x'}$.



One–line summary.

$$\boxed{\mathrm{Tr}_{L/K}(x) \;:=\; \mathrm{tr}\bigl(\, y \mapsto x \cdot y \,\bigr)}$$

the field trace of x is the categorical trace of "multiply by x."

5. What this project's trace is

In this formalisation the ground field is $K = \mathbb{F}_2 = \mathbb{Z}/2$ and L is a finite field of characteristic 2. Two facts pin everything down.

(i) Library trace = categorical trace. Mathlib's `Algebra.trace K L` is defined as exactly the map of §4:

$$\texttt{Algebra.trace}\ K\ L\ (x) \;=\; \mathrm{tr}(m_x) \;=\; \mathrm{Tr}_{L/K}(x).$$

(ii) Over $\mathbb{F}_2$ the loop becomes a Frobenius sum. For finite fields the trace has a closed form (`FiniteField.algebraMap_trace_eq_sum_pow`):

$$\mathrm{Tr}_{L/\mathbb{F}_2}(x) \;=\; \sum_{i=0}^{n-1} x^{2^i}, \qquad n = [L : \mathbb{F}_2].$$

The right–hand side is precisely the project's own trace function `Tr` / `truncTrace`. This equality is checked, `sorry`–free, in `Equation1/TraceLibrary.lean`:

$$\begin{tikzcd}
x \arrow[r, "m_x"] \arrow[dr, "\sum_{i<n} x^{2^i}"'] & \mathrm{tr}(m_x) = \texttt{Algebra.trace}\ \mathbb{F}_2\ L\ x \arrow[d, equal] \\
 & \texttt{Tr}\ n\ x = \texttt{truncTrace}\ n\ x
\end{tikzcd}$$

So the "trace" appearing throughout the equation–(1) development is not an ad–hoc sum: it is the categorical loop of §3 applied to the multiplication endomorphisms of §4, specialised to $\mathbb{F}_2$ where the loop unwinds into the Frobenius sum $\sum_i x^{2^i}$.

Diagrams: TikZ / tikz-cd. Rebuild with `tectonic TraceMap_Categorically.tex`.